
Fall 2026 Opening Set

UTSA Problem Solving Club

“Don’t just read it; fight it!” — Paul Halmos

The Chicken McNugget Problem

McDonald’s sells Chicken McNuggets in boxes of 6, 9, and 20. Using any combination of boxes, some numbers of nuggets can be bought exactly and some cannot (e.g., you cannot buy exactly 1, 2, or 43 nuggets, but you can buy 15, or 46).

Find the largest number of nuggets that cannot be bought exactly, and prove that every larger number can be.

A Fair Coin from a Biased One

You are given a coin that lands heads with some unknown probability p , where $0 < p < 1$ (it need not be fair, and you do not know p).

Describe a procedure, using only flips of this coin, that produces a single fair (50/50) coin flip — regardless of the value of p .

Five Points in a Square

Five points are placed anywhere inside (or on the boundary of) a square of side length 1.

Prove that some two of the five points are at distance at most $\frac{\sqrt{2}}{2}$ from each other.

Coprime Neighbors

Let n be a positive integer, and choose any $n + 1$ integers from the set $\{1, 2, 3, \dots, 2n\}$.

Prove that two of the chosen integers must be coprime (i.e., share no common factor other than 1).

The Two Envelopes Paradox

You are shown two identical envelopes and told that one contains exactly twice as much money as the other. You pick one at random and open it, finding an amount A inside. Before deciding whether to keep it, you reason as follows: “The other envelope contains either $A/2$ or $2A$, each equally likely, so its expected value is $\frac{1}{2} \cdot \frac{A}{2} + \frac{1}{2} \cdot 2A = \frac{5}{4}A$, which is more than A — so I should switch.”

But this argument never used the actual value of A : it would tell you to switch no matter what you found, even before you opened the envelope — which is absurd, since then you’d want to switch back, and then switch again, forever. **Find the flaw in the reasoning.**

Ants on a Stick

Some number of ants stand on a straight rod 1 meter long, each facing either left or right, each walking at a constant speed of 1 meter per minute. Whenever two ants meet head-on, they instantly reverse direction and continue walking (at the same speed) the other way. An ant that reaches an end of the rod falls off and is gone.

Prove that, no matter how many ants there are, where they start, or which way each one is initially facing, every ant has fallen off the rod within 1 minute.

The Counterfeit Coin

You have 12 coins that look identical, but exactly one is counterfeit: it weighs slightly differently from the other 11 (either heavier or lighter — you don't know which). You have a balance scale with no weights, which only tells you whether the two sides balance or which side is heavier.

Using the balance only 3 times, find the counterfeit coin and determine whether it is heavier or lighter than the rest.

The Josephus Survivor

n people stand in a circle, numbered 1 through n in order. Starting at person 1 and proceeding around the circle, every second remaining person is eliminated, until only one person is left.

Find, as a function of n , the starting position of the sole survivor.

Hint: Work out the survivor for $n = 1, 2, \dots, 8$ by hand, write n in binary, and look for a pattern.